



STORM: Segment, Track, and Object Re-Localization from a Single Image

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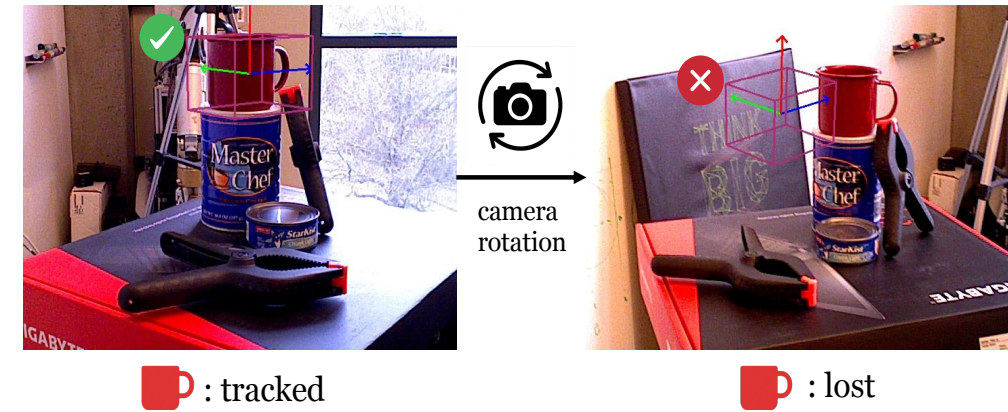


Code

Why 6D Tracking Fails in the Wild

Initial Input

New Input



☑ : tracked

☒ : lost

• Deployment remains labor-intensive.

Existing pipelines often rely on curated CAD models, manual masks, or per-object adaptation, limiting direct deployment when inference-time object models are unavailable.

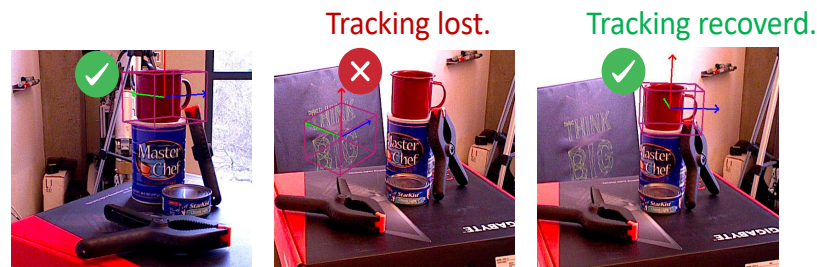
• Reference and query views are severely shifted.

A clean reference image must be matched to dynamic query observations affected by heavy occlusion, motion blur, and drastic viewpoint changes.

• Trackers fail silently.

Existing trackers often operate “blindly”: they lack a built-in tracking-validity signal to detect drift, making autonomous recovery difficult.

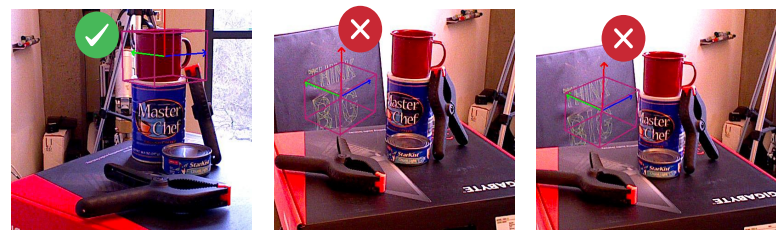
STORM



Tracking lost.

Tracking recover.

Ground Truth + Foundation Pose



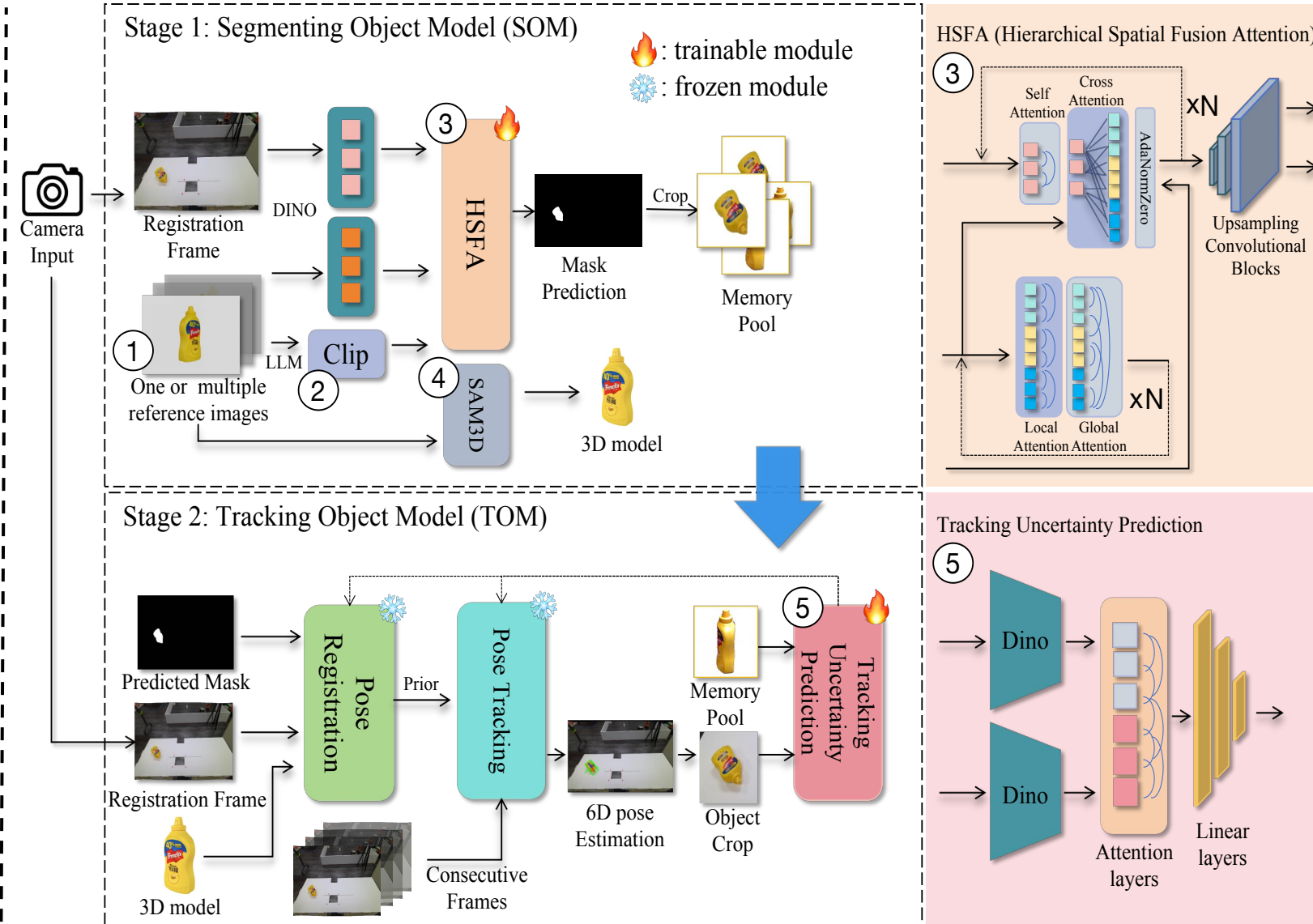
Tracking lost.

Tracking lost.



Time step

STORM: Reference-Guided Tracking with Self-Verification



• STORM couples segmentation and tracking in a closed loop.

Given one or multiple reference images, the Segmenting Object Module (SOM) predicts the target mask by aligning reference and query features through Hierarchical Spatial Fusion Attention (HSFA). The predicted mask and SAM3D-reconstructed geometry initialize downstream 6D pose registration and tracking.

• SOM: Hierarchical Reference-Query Fusion.

HSFA first aggregates reference-view evidence into an object-centric representation, then lets query features attend to the reference features through cross-attention. This replaces brittle template matching with learned feature fusion under clutter, occlusion, and viewpoint changes.

• TOM: Tracking-Loss Verification.

The Tracking Object Module compares the current observation crop with a memory pool of successful tracks. It predicts whether the current observation remains compatible with the tracked object and converts this compatibility into a continuous failure score.

• Re-localization Trigger.

A high failure score indicates low compatibility. If the smoothed score remains high for consecutive frames, STORM declares tracking loss and triggers automatic re-localization.

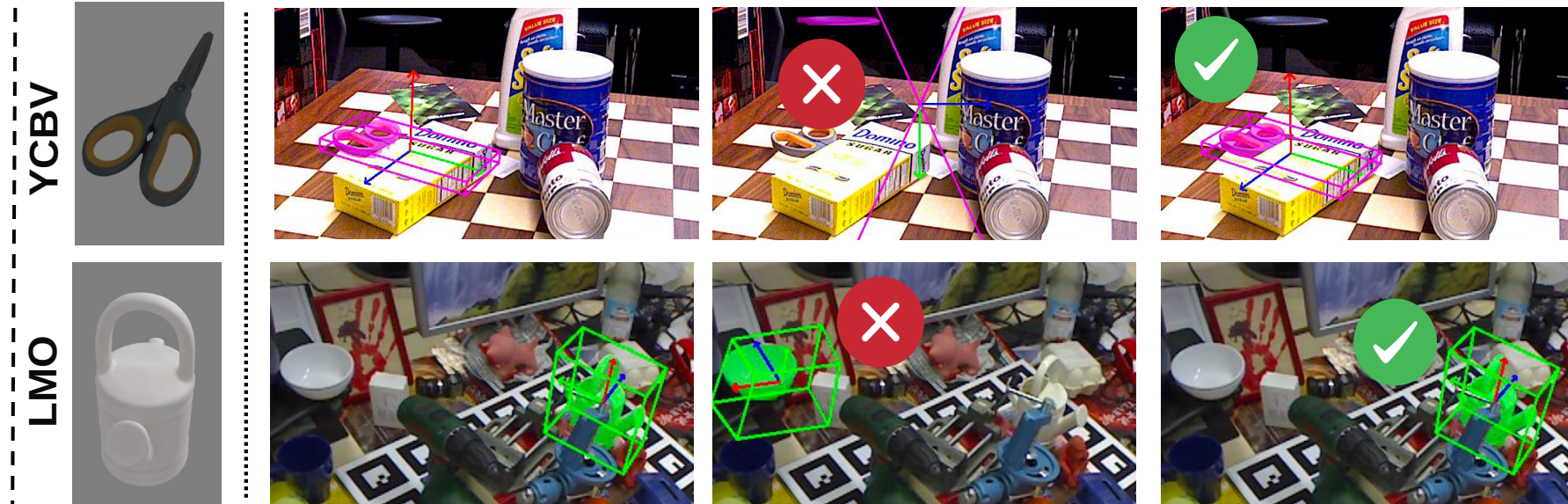
Robust Pose Tracking with Automatic Recovery

3D Model

Ground Truth

CNOS + FoundationPose

STORM (Ours)



1. STORM (SOM) achieves high-quality and efficient segmentation from a single reference image

	mAP (↑)	Annotation-free					Supervised	
		LM-O	T-LESS	TUD-L	HB	YCB-V	Mean ↑	Time (s)
Annotation / adaptation-free	STORM (SOM, Ours)	57.8±2.18	53.0±1.67	73.3±2.26	74.1±0.50	80.3±1.41	67.7±1.44	0.046
	NOCTIS	○48.9	47.9	58.3	60.7	○68.4	○56.8	0.990
	LDSeg	47.8	○48.8	○58.7	○62.2	64.7	56.4	1.890
	MUSE	47.8	45.1	56.5	59.7	67.2	55.3	0.559
	NIDS	43.9	○49.6	55.6	62.0	65.0	55.2	0.485
	Prisma-MPG-Complex	46.0	45.8	58.4	59.6	61.0	54.2	1.276
	SAM6D	46.0	45.1	56.9	59.3	60.5	53.6	2.795
	ViewInvDet	41.0	38.5	46.4	54.5	61.6	48.4	1.700
	CNOS (FastSAM)	39.7	37.4	48.0	51.1	59.9	47.2	○0.221
	CNOS (SAM)	39.6	39.7	39.1	48.0	59.5	45.2	1.847
Supervised	ZebraPoseSAT-EffNetB4	51.6	72.1	71.8	68.9	73.1	67.5	○0.080
	CosyPose	○37.5	○54.4	○48.9	○47.1	○52.0	○48.0	0.050
	Mask R-CNN	37.5	54.4	48.9	47.1	42.9	46.2	0.100

2. STORM improves annotation-free estimation

	Method	ADD _{AUC}	ADD-S _{AUC}	AR
LM-O	FP + CNOS	57.0	68.0	41.0
	STORM (Ours)	74.0±1.28	89.0±1.25	53.0±2.02
	FP + Ground Truth	78.0	93.0	56.0
YCB-V	FP + CNOS	73.0	92.0	69.0
	STORM (Ours)	77.0±1.25	98.0±1.20	73.0±1.23
	FP + Ground Truth	78.0	99.0	74.0

3. STORM automatically recovers from tracking failures

Method	ADD _{AUC}	ADD-S _{AUC}	AR ↑	Time(ms)
STORM	74.64±1.25	88.56±1.89	67.85±2.13	98±3
FP Tracking	52.76	66.76	50.09	84

4. SAM3D-generated 3D models provide reliable input for pose estimation

Method	VSD↑	IoU	ADD-S _{AUC}
SAM3D (unaligned) + FP	22.19±3.3	48.27±6.24	44.92±5.23
SAM3D (aligned) + FP	60.72±0.32	84.63±1.43	99.12±0.87
GT + FP	60.97	88.50	98.95

5. STORM gains performance by attention-based matching

Models	Acc. (%)	AUROC (%)	Time (ms)
Cosine Similarity	87.55	94.2	12.4
TOM (w/o Attn.)	95.23±0.26	95.6±0.16	12.2±0.30
TOM (Attn., 1L)	98.36±0.21	96.4±0.13	13.5±0.37
TOM (Attn., 2L)	97.84±0.14	96.2±0.10	16.8±0.43